

PORTFOLIO INTELLIGENCE PLATFORM

# AI Portfolio Rationalization Advisor

Centaur Governance Model · Q1 2026

## User Guide & Reference Documentation

Version 1.0 · May 2026 · Prepared by K. Turner, Senior AI Product Manager

### About This Document

The AI Portfolio Advisor is a fully functional enterprise AI governance platform implementing the Centaur model — AI recommends, humans decide — for application portfolio rationalization decisions. It is deployed live on Streamlit Community Cloud and operates with realistic enterprise portfolio data. This guide is intended for interviewers, business stakeholders, and technical reviewers evaluating the governance architecture, explainability design, and human-in-the-loop workflow.

| Item                            | Detail                                                                                  |
|---------------------------------|-----------------------------------------------------------------------------------------|
| Live URL                        | <a href="https://ai-app-advisor.streamlit.app">https://ai-app-advisor.streamlit.app</a> |
| Screens 1–5                     | Fully public — no login required                                                        |
| Screen 6 (Semantic CMDB Search) | Access key required — contact demo owner                                                |
| Data mode                       | Demo mode (CSV) - automatic fallback when ServiceNow PDI is unreachable                 |

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# 1 The Centaur Governance Model

The Centaur model is a human-AI collaboration framework for enterprise decision making. Named after the mythological figure combining human and horse, it pairs AI analytical capability with human judgment, accountability, and contextual reasoning.

## Core Principle

**Core Principle**

*AI recommends - humans decide. No AI output in this platform is final until reviewed and approved by a designated Application Portfolio Manager (APM) / Application Owner. Every recommendation includes an explainability layer showing exactly which signals drove the output.*

## The Three Governance Layers

| Layer          | Role                                                                                                            | Who Acts                |
|----------------|-----------------------------------------------------------------------------------------------------------------|-------------------------|
| AI Engine      | Analyzes 8 signals per application, generates recommendation with confidence score and Jagged Frontier position | Automated               |
| APM Review     | Validates recommendation against business context, approves, overrides with written rationale, or escalates     | APM / Application Owner |
| CIO Escalation | Reviews edge cases, high-stakes decisions, and escalations flagged by the APM                                   | Executive Sponsor       |

*Every decision, approval, override, or escalation, is captured in the Decision Audit Log with timestamp, reviewer identity, and written rationale. This creates a full governance trail.*

# 2 Design Principles

Every architectural and UX decision in this platform was driven by a governance principle, not a technical convenience. These four principles are the foundation of the Centaur model as implemented.

|                                              |                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Explainability before action</b>          | Every AI recommendation surfaces the signals that drove it before the APM / Application Owner is asked to approve or override. Users cannot act responsibly on outputs they cannot interrogate. The AI Analysis screen exists specifically to make black-box reasoning visible before it influences a decision.           |
| <b>Override must be easy and accountable</b> | Structural friction on override produces automation bias, users accept AI recommendations not because they agree but because disagreeing requires more effort. The Validation Queue makes override as easy as approval, while requiring written rationale to maintain accountability and enable future model improvement. |
| <b>Governance scales with confidence</b>     | The Jagged Frontier position tells the APM how much to trust the AI recommendation before reviewing it. Inside means high reliability, Edge                                                                                                                                                                               |

|  |                                                                                                                                                                                              |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | means validate carefully, Outside means exercise independent judgment. This calibrates oversight intensity to actual AI reliability rather than applying uniform scrutiny to every decision. |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                         |                                                                                                                                                                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Every decision is owned | The audit log captures reviewer identity, timestamp, and rationale for every action. Accountability stays with the designated APM or Application Owner, not the AI system itself. AI can support the decision, but ownership of the outcome remains human regardless of confidence score. |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3 Getting Started

#### Accessing the Application

The application is hosted on Streamlit Community Cloud and accessible via any modern web browser. No installation is required.

| Step | Action                                                                                              | Notes                              |
|------|-----------------------------------------------------------------------------------------------------|------------------------------------|
| 1    | Navigate to <a href="https://ai-app-advisor.streamlit.app">https://ai-app-advisor.streamlit.app</a> | Chrome, Firefox, Edge, or Safari   |
| 2    | Wait for the app to load (10–20 seconds on first visit)                                             | Free tier may sleep between visits |
| 3    | Use the left navigation panel to move between screens                                               | Screens 1–5 require no login       |
| 4    | For Screen 6, enter the access key when prompted                                                    | Contact demo owner for key         |

#### Navigation

The left sidebar contains the primary navigation. The currently active screen is highlighted in navy blue. The Validation Queue item displays a red dot indicator when pending decisions require APM review.

#### Data Mode Indicator

A status badge at the top of the sidebar indicates the current data source:

| Badge        | Meaning                                                                             |
|--------------|-------------------------------------------------------------------------------------|
| Yellow badge | Demo mode — CSV data loaded locally. All governance workflows are fully functional. |
| Green badge  | Live mode — Connected to ServiceNow PDI. Live CMDB data loaded.                     |

*For interview demonstrations, the app runs in Demo mode with realistic CSV data representing a 15-application enterprise portfolio. All metrics, recommendations, and governance workflows are fully functional in Demo mode.*

4 Screen 1 — Portfolio Overview

The Portfolio Overview is the primary dashboard, providing an executive-level summary of the application portfolio and its current rationalization cycle status.

Key Metrics

| Metric            | Description                                                                             |
|-------------------|-----------------------------------------------------------------------------------------|
| At-Risk Apps      | Applications flagged for immediate attention based on combined risk signals             |
| Portfolio TCO     | Total Cost of Ownership across all 15 applications in the current cycle                 |
| Pending Decisions | Applications awaiting APM review before the cycle close date                            |
| Cycle Accuracy    | Percentage of AI recommendations validated as correct by APM / Application Owner review |

Filters

Three multi-select filter bars allow slicing the portfolio by Lifecycle stage, AI Recommendation type, and Business Unit. All filters interact in real time, the application table updates immediately on every selection change.

Application Table

The table lists all applications matching the active filters with the following columns:

- Application
- Business Unit
- Criticality
- Lifecycle
- Annual Cost
- Incidents (12M)
- Tech Debt
- AI Recommendation
- Status

5 Screen 2 — AI Analysis & Signals

This screen provides full explainability for every AI recommendation. Select any application from the dropdown to see the complete signal breakdown that drove the AI output.

Application Summary Card

The top card shows the application's key attributes - business unit, lifecycle stage, annual cost, and incident volume - alongside the AI recommendation badge and confidence score.

Frontier Position

Each application is classified against the Jagged Frontier, a concept describing where AI performs reliably versus where human judgment is essential:

| Position | Meaning                                       | Action Required                    |
|----------|-----------------------------------------------|------------------------------------|
| Inside   | Strong data signals, high AI reliability      | Can approve with confidence        |
| Edge     | Competing signals reduce reliability          | Should validate carefully          |
| Outside  | Insufficient data for reliable recommendation | Must exercise independent judgment |

Signal Breakdown — 8 Signals

A horizontal bar chart displays the eight signals used by the AI engine, each scored 0–100 with a severity badge (High/Medium/Low):

| Signal                       | Description                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------|
| Incident trend               | 12-month (12m) incident direction and trajectory, is the rate rising or declining?                    |
| Duplicate functionality      | Overlap with other portfolio applications                                                             |
| Technical debt index         | Code quality, age, and maintenance burden score                                                       |
| Lifecycle status             | Current / Aging / End-of-Life classification                                                          |
| 12-month incident volume     | Raw incident count over the trailing year                                                             |
| Stability inverse            | Inverse of application stability score                                                                |
| Replacement candidate exists | Whether a viable replacement has been identified                                                      |
| Critical CVEs                | Count of critical security vulnerabilities from vulnerability records (scored 0–4 in current dataset) |

The 'Why this recommendation?' panel on the right provides a plain-English explanation of the top contributing signals, making the AI reasoning accessible to non-technical stakeholders.

6 Screen 3 — Validation Queue

The Validation Queue is the APM's primary workspace. It lists all applications with AI recommendations that have not yet been reviewed in the current cycle.

Reviewer Identity

The reviewer's name field at the top of the screen captures the APM identity. This name is stamped on every decision made during the session and recorded in the audit log.

## Decision Actions

| Action          | When to Use                                                            | Audit Outcome                                           |
|-----------------|------------------------------------------------------------------------|---------------------------------------------------------|
| Approve         | AI recommendation is correct and appropriate given business context    | Logged as Approved with reviewer identity and timestamp |
| Override        | APM disagrees with AI recommendation - must provide written rationale  | Logged as Overridden with rationale captured            |
| Escalate to CIO | Decision requires executive input - high risk, high cost, or ambiguous | Logged as Escalated, flagged for CIO review             |

### Governance Control

*Override requires a written rationale of at least one sentence. This is a deliberate governance control - it forces the APM to articulate their reasoning, creating accountability and enabling future model improvement.*

## Recommendation Variance Alert

If an AI recommendation changes from the previous review cycle, a variance alert is displayed on the application card. This flags potential LLM non-determinism, where the same inputs may produce different outputs across review cycles, and indicates that additional scrutiny is required before approval. All variance alerts are automatically escalated under governance policy. Because non-deterministic output increases the risk of misapplication or inconsistent decision support, recommendations should not be approved without careful APM review.

## 7 Screen 4 — Decision Audit Log

The Decision Audit Log provides a complete, immutable record of every governance decision made in the current cycle. It is the primary accountability mechanism of the Centaur model.

## Log Entries

Each audit entry captures the following:

- Application name and ID
- Decision type (Approved / Overridden / Escalated)
- AI recommendation at time of decision
- Reviewer name and timestamp
- Written rationale (for overrides)
- Confidence score at time of review

### Production Note

*The audit log is session-based in Demo mode. In a production deployment, entries would persist to a database and be exportable for compliance reporting.*

## Override Analysis

A summary panel shows the override rate for the current cycle, the percentage of AI recommendations that the APM disagreed with. A high override rate indicates output variance or data quality issues requiring investigation.

## 8 Screen 5 — Governance Dashboard

The Governance Dashboard provides portfolio-level analytics on the rationalization cycle, designed for senior stakeholder reporting and model performance monitoring.

### Cycle Summary

The top section shows cycle-level KPIs including total decisions made, approval rate, override rate, escalation rate, and overall cycle accuracy against historical benchmarks.

### SQDC Metrics — Portfolio Impact

The dashboard tracks portfolio health through the SQDC framework (Safety, Quality, Delivery, and Cost) measured run-over-run to show cycle-over-cycle improvement. These metrics connect AI governance outcomes directly to the operational performance framework used across industrial enterprises.

| Dimension | What It Measures                                                            |
|-----------|-----------------------------------------------------------------------------|
| Safety    | Regulatory risk applications identified and queued for retirement           |
| Quality   | True positive rate improving cycle-over-cycle; override rate declining      |
| Delivery  | Cycle review time reduction — from 6 weeks to 3 weeks in demonstration data |
| Cost      | Portfolio TCO optimization through retire and modernize decisions           |

### Recommendation Distribution

A breakdown of AI recommendations across the five categories:

- **Retain** — Application is performing well, no action required
- **Modernize** — Application requires investment to remain viable
- **Retire** — Application should be decommissioned
- **Replace** — Application should be replaced with an alternative
- **Evaluate** — Insufficient data — further assessment required

### Frontier Analysis

A chart showing the distribution of applications across Inside, Edge, and Outside frontier positions — providing insight into where the AI model is most and least reliable across the current portfolio.

## 9 Screen 6 — Semantic CMDB Search

### Access Required

*Screen 6 requires an access key. Contact the demo owner to receive the key before your session. The key is valid for the duration of your browser session.*

The Semantic CMDB Search is an AI-powered conversational interface that allows natural language queries against the full portfolio dataset. It is powered by Claude (Anthropic) via the Anthropic API.

### Access Gate

On first visit, Screen 6 displays an access key prompt. Enter the key provided by the demo owner. Once authenticated, the gate does not reappear for the remainder of your browser session.

### Query Quota

Each session is limited to 20 AI queries to manage API costs. A quota counter in the top-right corner shows remaining queries. The counter resets when you open a new browser session or clear your browser data.

### Suggested Prompts

Eight suggested prompts are displayed when the conversation is empty. Click any prompt to run it immediately:

- Which apps should we retire first and in what order?
- What's the total cost of all End of Life applications?
- Show me all high-criticality apps that are aging or end of life
- Which Finance apps carry the most security risk?
- Explain the dependency risk if we retire LegacyDB
- Compare ERP Core and DataWarehouse — which is a higher priority to modernize?
- What would the annual savings be if we retired all recommended apps?
- Which apps are most likely to have an outage in the next 6 months?

### Portfolio Quick Reference

An expandable panel at the bottom of the screen shows a quick reference table of all 15 applications with their key attributes — useful for formulating precise queries.



## 10 Data Sources & Demo Mode

The application supports two data modes that are detected automatically at startup.

### Demo Mode (CSV)

When no ServiceNow credentials are configured or the PDI is unreachable, the application loads from five CSV files representing a realistic 15-application enterprise portfolio. All screens are fully functional in Demo mode.

| File                          | Contents                                 |
|-------------------------------|------------------------------------------|
| 01_cmdb_ci_business_app.csv   | Application inventory with metadata      |
| 02_cmdb_rel_ci.csv            | Application dependency relationships     |
| 03_sn_vul_vulnerable_item.csv | Security vulnerability records           |
| 04_business_capability.csv    | Business capability mappings             |
| 05_incident_summary.csv       | 12-month incident summary by application |

### Live Mode (ServiceNow)

When ServiceNow credentials are present in the application secrets, the app attempts to connect to a ServiceNow Personal Developer Instance (PDI) at startup. If successful, the sidebar badge changes to green and live CMDB data is loaded. If the connection fails, the app silently falls back to CSV data.

**Production Path**

*The CSV data schema maps directly to ServiceNow CMDB fields. Connecting to a live ServiceNow instance is a data integration task, not an architectural one. The governance layer, AI recommendation engine, validation queue, and audit trail are all fully production-grade as built.*

## 11 FAQ & Troubleshooting

**Q: How does this differ from a standard AI recommendation tool?**

A: Most AI tools surface recommendations without governance infrastructure. The AI Portfolio Advisor implements a complete Centaur governance layer — every recommendation goes through human validation before becoming a decision, every decision is audited with reviewer identity and written rationale, and the system explicitly surfaces where AI reliability is lower so human judgment is applied proportionally. The SQDC dashboard connects governance outcomes directly to operational performance metrics.

**Q: What would a production deployment look like?**

A: The architecture is designed for production. The ServiceNow integration replaces the CSV data layer without changing any governance logic. The session-based audit log would persist to a database. The access control on Screen 6 would integrate with enterprise SSO. The governance dashboard would feed into existing reporting workflows. The governance layer — the part that matters most — is fully production-grade as built.

**Q: The app is taking a long time to load.**

A: Streamlit Community Cloud free tier apps sleep after inactivity. The first load after a sleep period takes 15–20 seconds. This is normal — wait for the spinner to complete and the app will load fully.

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**Q: Screen 6 shows an access key prompt.**

A: Screen 6 requires an access key to control API usage costs. Contact the demo owner (K. Turner) to receive the key. Enter it in the prompt and press Enter.

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**Q: I've used all 20 queries on Screen 6.**

A: The per-session query limit has been reached. Open a new browser tab or window to start a fresh session with a new quota.

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**Q: The sidebar shows 'Demo mode · CSV data'.**

A: This is expected behavior. The app is running with realistic CSV data. All screens are fully functional — the demo mode label indicates data source only.

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**Q: A decision I made isn't showing in the Audit Log.**

A: The audit log is session-based in Demo mode. If you refreshed the browser, session state is reset. In production, decisions would persist to a database.

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**Q: The AI recommendation seems wrong for a specific application.**

A: This is intentional — some applications are placed at the Edge or Outside frontier where AI reliability is lower. Use Screen 2 to review the signal breakdown and exercise APM judgment on the Validation Queue.

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**Q: Can I export the portfolio data or decisions?**

A: Export functionality is not included in the Demo version. In a production deployment, CSV export and PDF reporting would be standard features.

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**Contact**

*For additional questions or to schedule a live walkthrough, contact K. Turner, Senior AI Product Manager. Live application: <https://ai-app-advisor.streamlit.app>*